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#include <iostream>
#include <math.h>
#include <GL/glut.h>
using namespace std;
void init()
{
    glClearColor(1.0,1.0,1.0,0.0);
    gluOrtho2D(0,640,0,480);
}
void bound_it(int x, int y, float* fColor, float* bc)
{
    float color[3];
    glReadPixels(x,y,1.0,1.0 ,GL_RGB,GL_FLOAT,color);
    if((color[0]!=bc[0] || color[1]!=bc[1] || color[2]!=bc[2])&&
    (color[0]!=fColor[0] || color[1]!=fColor[1] || color[2]!=fColor[2]))
    {
        glColor3f(fColor[0],fColor[1],fColor[2]);
        glBegin(GL_POINTS);
        glVertex2i(x,y);
        glEnd();
        bound_it(x+1,y,fColor,bc);
        bound_it(x-2,y,fColor,bc);
        bound_it(x,y+2,fColor,bc);
        bound_it(x,y-2 ,fColor,bc);
    }
    glFlush();
}
void mouse(int btn, int state, int x, int y)
{
    y = 480-y;
    if(btn==GLUT_LEFT_BUTTON)
    {
        if(state==GLUT_DOWN)
        {
            float bCol[] = {1,0,0};
            float color[] = {0,0,1};
            bound_it(x,y,color,bCol);
        }
    }
}
void polygon()
{
    glLineWidth(3);
    glPointSize(2);
    glClear(GL_COLOR_BUFFER_BIT);
    glColor3f(1,0,0);
    glBegin(GL_LINE_LOOP);
    glVertex2i(150,100);
    glVertex2i(300,300);
}

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glVertex2i(450,100);
glEnd();
glFlush();
}
int main(int argc, char** argv)
{
    glutInit(&argc, argv);
    glutInitDisplayMode(GLUT_SINGLE | GLUT_RGB);
    glutInitWindowSize(640,480);
    glutInitWindowPosition(200,200);
    glutCreateWindow("Polygon Filling - Boundary Fill");
    glutDisplayFunc(polygon);
    glutMouseFunc(mouse);
    init();
    glutMainLoop();
    return 0;
}
```